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Research paper



UAV-USV heterogeneous collaboration collision avoidance decision-making based on multi-agent reinforcement learning

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ABSTRACT

This study addresses a significant gap in the application of multi-agent reinforcement learning (MARL) to unmanned aerial vehicle and unmanned surface vehicle (UAV-USV) heterogeneous collaboration collision avoidance decision-making (HCCAD). The MARL has demonstrated considerable promise in homogeneous multi-agent systems; however, its extension to heterogeneous platforms with disparate dynamics, sensing modalities, and action spaces remains underexplored. To bridge this gap, an optimal baseline multi-agent proximal policy optimisation (OB-MAPPO) framework tailored for HCCAD problem is proposed. The innovations of the proposed method are: (1) The framework incorporates the advantage decomposition mechanism and optimal baseline to reduce gradient variance and mitigate redundant parameter sharing caused by heterogeneous action spaces. (2) A multimodal fusion network based on the Swin Transformer is introduced to effectively integrate visual data from the UAV and LiDAR observations from the USV, resolving spatial discrepancies in cross-domain perception. (3) The reward function is designed to incorporate collaborative error, ship domain, and immediate danger situations. Simulations show our method achieves superior convergence, success rate, and generalization in complex dynamic environments compared to existing MARL and conventional methods. Our work presents a novel, robust, and scalable paradigm for collision avoidance in cross-domain operations involving both aerial and maritime vehicles.

1. Introduction

In recent years, the increasing complexity and risk associated with global maritime activities have driven the development of unmanned systems toward heterogeneous collaboration. The unmanned aerial vehicles (UAVs) provide large-scale situational awareness through their high-altitude manoeuvrability (Zhang et al., 2025), while unmanned surface vehicles (USVs) perform long-duration operational tasks by leveraging their endurance on the water's surface (Xia et al., 2023). The integrated "air-sea" three-dimensional system formed by combining UAVs and USVs has shown significant advantages in maritime search and rescue (Zhang et al., 2022), ship pilotage (Liu et al., 2024), and military reconnaissance missions (Wang et al., 2023b). This cross-domain decision-making effectively addresses the limitations of individual platforms in terms of perception range, task efficiency, and adaptability to dynamic environments. As the foundation for "air-sea" system, UAV-USV heterogeneous collaboration collision avoidance decision-making (HCCAD) not only improves the efficiency and safety of maritime missions but also constitutes a critical technical challenge that demands urgent resolution.

As a complex heterogeneous decision-making problem, the primary objective of UAV-USV HCCAD is to achieve autonomous collision avoidance in unknown dynamic environments by integrating UAV-based visual assistance with USV's LiDAR observations, while maintaining coordinated formation between the UAV and USV. However, research in this field still faces numerous challenges. Unlike isomorphic agents, heterogeneous agents face challenges in integrating differences in their observation spaces arising from sensors of varying dimensions. Moreover, it is worthwhile to further explore how HCCAD can enable UAV-USV systems to achieve dynamic decision-making with strong generalisation capabilities, while maintaining high-precision collaboration.

In the research on UAV-USV collaboration, integrating computer vision with model predictive control (MPC) and event-triggered control

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(ETC) is widely recognised as an effective and commonly adopted approach (Zhang et al., 2024, 2025). Such methods acquire posture data from visual information and pre-compensate for the effects of interfering factors, substantially reducing response latency. Moreover, they trigger optimisation calculations only when significant environmental changes occur, thereby minimising computational load and improving overall system efficiency. Chen et al. (2025) developed a saturation control strategy to address the guidance problem of UAV and introduced a performance compensation control scheme within the kinetic energy loop to enhance the robustness of the UAV-USV control system. To achieve path convergence in UAV control, Li et al. (2023) combined the dynamic surface control (DSC) technique with the backstepping method and applied nonlinear decoupling to determine the required roll and pitch angles for the UAV. Li et al. (2024c), present a bidirectional communication topology network, in which an extended Kalman filter (EKF) is implemented on each individual UAV and USV to mitigate the effects of unknown disturbances. The integration of computer vision allows the system to acquire environmental information more effectively and facilitates cross-platform situational awareness sharing (Li et al., 2016; Wu et al., 2024). Huang et al. (2022) proposed an extended visual servo framework that integrates nonlinear differentiators and smooth saturation models into the feedback control system and implements hierarchical control for heterogeneous decoupling. Ji et al. (2022), by integrating occlusion perception with topological structure, ensure the UAV's dynamic optimised trajectory, thereby reducing visual tracking errors. Wang et al. (2024) divide the UAV-USV search mission into two subtasks. First, a mapping model is established to identify optimal waypoints and balance the workload between the UAV and USV. Then, by combining waypoint clustering with closed-loop gain control, the search trajectories for each agent are generated. However, the aforementioned methods still exhibit certain limitations, which are primarily reflected in the following aspects: (1) a high dependency on complex heterogeneous kinematic models; (2) the visual strategy fails to ensure high-precision collaboration error; (3) insufficient adaptability and generalisation capability when handling complex environments.

For the USV collision avoidance problem, traditional approaches typically integrate global and local path planning methods, such as risk assessment strategies, population-based algorithms, and artificial potential field (APF) algorithms. Xiao et al. (2025), construct an adaptive function based on individual waypoints, and the search traversal process is enhanced through a chaotic multi-group parallel optimisation strategy to improve navigation safety. Wan et al. (2025) develop a cooperative game model for multiple USVs incorporating an accelerated convergence update mechanism based on game optimisation. This enhancement improves the performance of the particle swarm optimisation (PSO) algorithm and ensures USV behaviour complies with COLREGs (Perera et al., 2011; Zhang et al., 2015). For APF and its derivative algorithms, scholars commonly employ methods such as modelling based on harmonic functions (Jadhav et al., 2023), combining the ETC with the repulsive force function (Li et al., 2024a), and integrating with heuristic algorithms (Pongsirijinda et al., 2024) to improve decision-making efficiency. The aforementioned methods have exhibited stable performance in static and low-dynamic environments. Nevertheless, their effectiveness diminishes in complex, highly dynamic settings and high-dimensional action spaces, where they are prone to issues such as local optima and slow convergence rates (Zhang et al., 2022; Liu et al., 2022).

In recent years, reinforcement learning (RL) has exhibited remarkable performance in adaptability and generalisation when addressing behavioural decision-making problems, offering innovative solutions to aforementioned challenges. Extensive work has demonstrated that collision-free decision-making can be effectively achieved by employing strategies such as optimizing the reward function (Cui et al., 2024), constructing a hierarchical framework (Zhu and Hayashibe, 2023), and applying the expert paradigm for inverse inference (Cao et al., 2022). The integration of state space with sensors (e.g., LiDAR and cameras) reduces reliance on specific observable data while improving the generalisation capability of RL. Research has demonstrated that network models trained through RL can be effectively deployed on real-world USVs and UAVs, enabling autonomous decision-making in actual environments (Lou et al., 2025). Furthermore, it has also been found that incorporating specialised neural network architectures (e.g., long short-term memory (LSTM) and gated recurrent unit (GRU)) into RL frameworks can effectively capture and utilize critical historical information. This allows the system to retain relevant past states, thereby improving the accuracy of predicting subsequent state spaces.

Shen et al. (2019) combined the deep Q-learning (DQN) with the ship domain, achieving the first real-world intelligent collision avoidance for three USVs, thereby laying a crucial foundation for the application of RL in solving the HCCAD problem. The DQN and its variants (e.g., double DQN and duelling DQN) have gained widespread use due to their high sample efficiency and robust training stability. However, value-based RL faces the intractable problem of overestimation, where uneven noise can excessively inflate action values. This phenomenon can result in erroneous action selection by the algorithm in complex environments. Lou et al. (2024), by incorporating the velocity obstacle (VO) and the dynamic window algorithm (DWA) into the design of the reward function, effectively mitigated the severe sparsity of reward signals in deep deterministic policy gradient (DDPG). Cui et al. (2025a) integrated the intention-sharing mechanism and attention communication, thereby reducing the communication cost among USV swarms within the multi-agent DDPG (MADDPG) framework. In multi-agent task allocation, Mao et al. (2025) proposed a strategy network with an encoder-decoder structure based on the attention mechanism, aiming to maximise the overall task value under the constraint of UAV flight distance. However, in multi-agent RL (MARL), both the state and action spaces grow exponentially as the number of agents increases, which leads to the so-called curse of dimensionality. Moreover, each agent has access only to localised environmental information, necessitating the ability to infer the states and intentions of others.

The emergence of multimodal feature fusion has effectively overcome the limitations associated with individual sensors, thereby promoting the development of a collaborative and collision avoidance integrated framework. Wang et al. (2023b) developed a UAV-USV search and rescue system based on visual input. This approach extracts positional information from images captured by the UAV and enhances the accuracy of visual positioning through the application of a convolutional neural network (CNN) and spatial graph structures. Cui et al. (2025b), integrated a Transformer architecture with proximal policy optimisation (PPO) to separately process continuous radar observations and discrete state information. Cui et al. (2023b) applied the Transformer in multimodal fusion, which has the potential to identify a large number of key points in regions with low feature representation. However, minimising the collaboration errors between UAV and USV to the greatest extent while ensuring the flexibility of the collision avoidance decision-making strategy remains an important and urgent issue in current research.

To address these limitations, this study proposes a UAV-USV HCCAD strategy based on the optimal baseline multi-agent PPO (OB-MAPPO) algorithm. The main contributions are summarised as follows.

(1) An integrated training framework based on OB-MAPPO is developed for collaboration and collision-avoidance tasks. By incorporating OB and advantage decomposition, it reduces policy exploration variance and mitigates redundant parameter sharing caused by heterogeneous action spaces.
(2) In the network architecture, the Swin Transformer module is integrated into the generalised multimodal fusion strategy, facilitating effective feature interaction between image and LiDAR data, thereby addressing the issue of spatial discrepancies in observations from heterogeneous agents.

(3) The reward function comprehensively incorporates multiple factors, including collaborative error, ship domain, and immediate danger situations. Then, simulations show that our method converges faster and generalises better than existing MARL and conventional strategies.

The rest of this paper is organised as follows: Section II presents the theoretical preliminaries of the model and algorithm. The proposed HCCAD strategy is introduced in Section III. Finally, the simulation is arranged in Section IV, and the conclusion is provided in Section V.

2. Introduction preliminaries and problem formulation

2.1. UAV-USV motion mathematical model

To ensure that the HCCAD model acquired after training exhibits the motion characteristics of both UAV and USV, this paper introduces a nonlinear mathematical model by Euler-Lagrange equations (Labbadi and Cherkaoui, 2019; Li et al., 2024b), which is described in Fig. 1, and with its corresponding equation can be established as:

$$\begin{cases} \dot{x}_s = u_s\cos(\psi_s) - v_s\sin(\psi_s) \\ \dot{y}_s = u_s\sin(\psi_s) + v_s\cos(\psi_s) \\ \dot{\psi}_s = r_s \\ m_{su}\dot{u}_s = f_u(\nu_s) + \tau_u + d_{wu} \\ m_{sv}\dot{v}_s = f_v(\nu_s) + d_{wv} \\ m_{sr}\dot{r}_s = f_r(\nu_s) + \tau_r + d_{wr}, \end{cases} \tag{1}$$

$$\begin{cases} m_a\dot{u}_{ax} = f_x(\nu_a) + R_x\tau_f + d_x \\ m_a\dot{u}_{ay} = f_y(\nu_a) + R_y\tau_f + d_y \\ m_a\dot{u}_{az} = f_z(\nu_a) - g_a + R_z\tau_f + d_z \\ \dot{r}_\phi = f_\phi(\nu_a) + \dfrac{d}{I_{xx}}\tau_\phi + d_\phi \\ \dot{r}_\theta = f_\theta(\nu_a) + \dfrac{d}{I_{yy}}\tau_\theta + d_\theta \\ \dot{r}_\psi = f_\psi(\nu_a) + \dfrac{d}{I_{zz}}\tau_\psi + d_\psi, \end{cases} \tag{2}$$

with,

$$\begin{cases} R_x = \cos(\phi_a)\sin(\theta_a)\cos(\psi_a) + \sin(\theta_a)\sin(\psi_a) \\ R_y = \cos(\phi_a)\sin(\theta_a)\cos(\psi_a) - \sin(\theta_a)\sin(\psi_a) \\ R_z = \cos(\phi_a)\cos(\psi_a), \end{cases} \tag{3}$$

where $O - X_{OG}Y_{OG}Z_{OG}$ represent UAV-USV inertial coordinate system, $O_S - X_SY_S$ is the ship body coordinate system, and $O_A - X_AY_AZ_A$ is the coordinate system of UAV. $X_S = [x_s, y_s, \psi_s]^{\mathrm{T}}$ are the position coordinates and the heading angle of the USV and $X_A = [x_a, y_a, z_a, \phi_a, \theta_a, \psi_a]^{\mathrm{T}}$ is the position coordinates, heading angle, roll angle and pitch angle of the UAV. $\boldsymbol{\nu}_s = [u_s, v_s, r_s]^{\mathrm{T}}$ represent the surge, sway and yaw speed of the USV. $f(\boldsymbol{\nu}_s) = [f_u, f_v, f_r]^{\mathrm{T}}$ are the uncertainties of USV. $\boldsymbol{\nu}_a = [u_{ax}, u_{ay}, u_{az}, r_\phi, r_\theta, r_\psi]^{\mathrm{T}}$ are the velocities in earth fixed coordinate, yaw, roll and pitch speed of the UAV. $f(\boldsymbol{\nu}_a) = [f_x, f_x, f_x, f_\phi, f_\theta, f_\psi]^{\mathrm{T}}$ denotes the uncertainties of the UAV. $d(\boldsymbol{\nu}_s) = [d_{wu}, d_{wv}, d_{wr}]^{\mathrm{T}}$ are the wind and waves disturbances of the USV. $d(\boldsymbol{\nu}_a) = [d_x, d_y, d_z, d_\phi, d_\theta, d_\psi]^{\mathrm{T}}$ are the disturbances of the UAV. $R_A = \left[R_x, R_y, R_z, \frac{d}{I_{xx}}, \frac{d}{I_{yy}}, \frac{d}{I_{zz}}\right]^{\mathrm{T}}$ denotes the gain matrix of the UAV which has been discussed by (Li et al., 2024). g_a is the gravitational acceleration. $\tau(\boldsymbol{\nu}_s) = [\tau_u, 0, \tau_r]^{\mathrm{T}}$ and $\tau(\boldsymbol{\nu}_a) = [\tau_f, \tau_\phi, \tau_\theta, \tau_\psi]^{\mathrm{T}}$ are the control forces/moments for USV and UAV, which can be respectively mapped to the propeller speed n_s and rudder angle δ_s of the USV, and the four-axis propeller velocity $\boldsymbol{w}_a = [w_1, w_2, w_3, w_4]^{\mathrm{T}}$ and steering angle δ_a of the UAV.

2.2. Visual positioning model

As an airborne perception platform, the UAV captures continuous image frames of the surrounding area of the USV using high-definition cameras. To generate a digital image, a camera model is required to project dense 3D points onto a 2D plane. The corresponding models of the UAV, USV, and camera are illustrated in Fig. 2. It is assumed that the camera coordinate system is aligned with that of the UAV, and that the camera centre coincides with the geometric centre of the UAV. Through rigid body transformation, 3D spatial points can be projected onto the camera coordinate system, which can be formulated as (4):

$$\begin{bmatrix} x_{cam} \\ y_{cam} \\ z_{cam} \end{bmatrix} = \boldsymbol{R}(\phi_{cam}, \theta_{cam}, \psi_{cam}) \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \boldsymbol{T}_{cam}, \tag{4}$$

where $(x_{cam}, y_{cam}, z_{cam})$ denotes a point coordinate within the camera coordinate system, $\boldsymbol{R}$ is the rotation matrix that transforms coordinates between the two systems, ϕ_{cam}, θ_{cam}, and ψ_{cam} denote the camera's rotation angles along the 3DOF, and $\boldsymbol{T}_{cam}$ represents the displacement vector between the two coordinate systems.

Under perspective projection, points in the 3D camera coordinate system are projected onto the 2D image sensor coordinate system. Without considering the effects of lens distortion, the projected coordinates are obtained:

$$\begin{cases} x_{im} = (f_{cam}\cdot x_c/z_c)/\rho_u + c_u \\ y_{im} = (f_{cam}\cdot y_c/z_c)/\rho_v + c_v, \end{cases} \tag{5}$$

where f_{cam} represents the focal length, whereas ρ_u and ρ_v are the horizontal and vertical pixel dimensions, respectively. c_u and c_v are the optical coordinate centres expressed in pixel units.

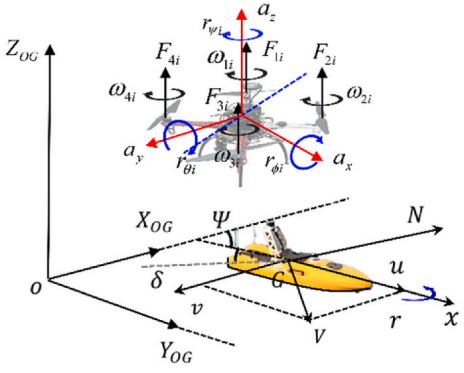


Fig. 1. The UAV-USV motion mathematical model.

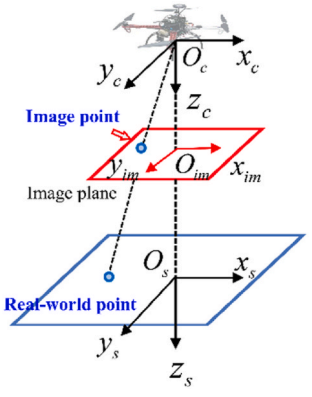


Fig. 2. The visual localization relevant model by UAV, USV, and camera.

2.3. COLREGs and ship domain model

The 13th to 15th articles of the COLREGs clearly stipulate the various situations involving vessel encounters and the corresponding responsibilities. These articles primarily address three scenarios: head-on, overtaking, and crossing situations, which are summarised in Fig. 3. Furthermore, to precisely define the minimum distance that the USV must maintain from the target ship (TS) and the appropriate timing for collision avoidance in immediate danger situation, a fuzzy quaternion ship domain (FQSD) model is adopted (Wang, 2010; Silveira et al., 2022), which is described in Fig. 4, and with its corresponding equation can be established as.

$$FQSD_r = \{(x,y)|f_r(x,y;Q_{FQSD}(r)) \le 1, Q_{FQSD}(r) = [R_{fore}, R_{aft}, R_{starb}, R_{port}], 0<r<1\}, \tag{6}$$

$$f_F(x,y;FQSD_r) = \left(\frac{2x}{[1+\mathrm{sgn}(x)]R_{fore} - [1+\mathrm{sgn}(x)]R_{aft}}\right)^2 + \left(\frac{2y}{[1+\mathrm{sgn}(y)]R_{starb} - [1+\mathrm{sgn}(x)]R_{port}}\right)^2, \tag{7}$$

$$R_i(r) = \left(\frac{\ln\frac{1}{r}}{\ln\frac{1}{r_0}}\right)^{\frac{1}{k}} R_i, i \in \{fore, aft, starb, port\}, \tag{8}$$

where r is the collision risk characterizing the corresponding fuzzy boundary of the FQSD, R_{fore} and R_{aft} is the longitudinal radius, R_{starb} and R_{port} represent lateral radius of the FQSD in starboard and port directions respectively, which can be calculated as:

$$\begin{cases} R_{fore} = \left(1 + 1.34\sqrt{k_{AD}^2 + (k_{DT}/2)^2}\right)\cdot L_{OS} \\ R_{aft} = \left(1 + 0.67\sqrt{k_{AD}^2 + (k_{DT}/2)^2}\right)\cdot L_{OS} \\ R_{starb} = (0.2 + k_{DT})\cdot L_{OS} \\ R_{port} = (0.2 + 0.75k_{DT})\cdot L_{OS}, \end{cases} \tag{9}$$

where L_{OS} is the length of own ship (OS), k_{AD} and k_{DT} are gains of the advance A_D and the tactical diameter D_T respectively, which can be established as:

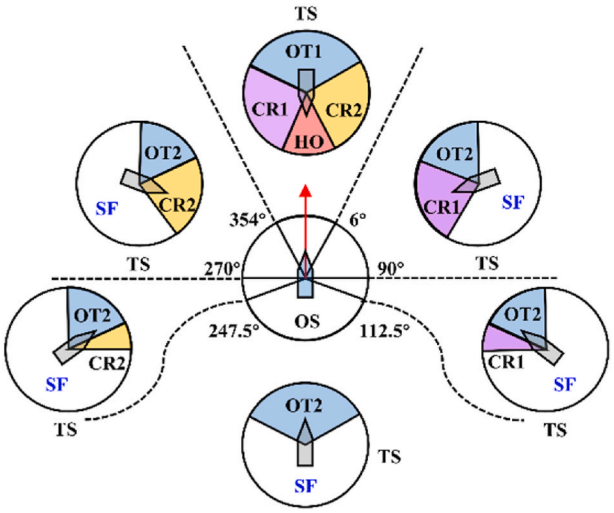


Fig. 3. The model of two-ship encounters situations for clauses 13th~15th in Chapter 2 of COLREGs.

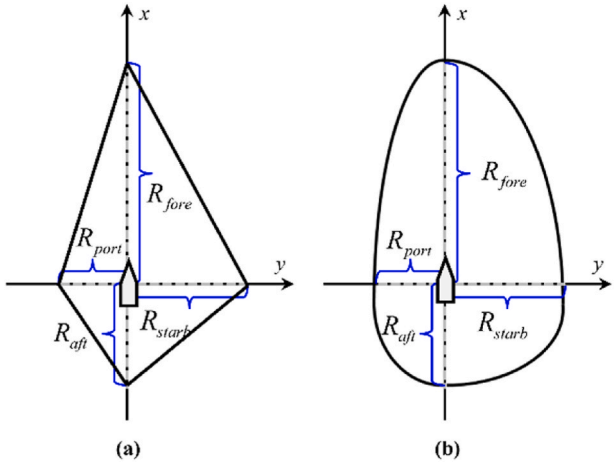


Fig. 4. The model of the FQSD ship domain.

$$\begin{cases} k_{AD} = A_D/L_{OS} = 10^{0.3591\ \lg V_{OS}+0.0952} \\ k_{DT} = D_T/L_{OS} = 10^{0.5441\ \lg V_{OS}-0.0795}, \end{cases} \tag{10}$$

where V_{OS} is the OS's speed. When a TS violates into the FQSD of the USV, it signifies that the USV is exposed to an immediate danger situation. Under such circumstances, to ensure safe navigation, the USV is permitted to deviate from the COLREGs.

3. Proposed approach

3.1. Multi-agent reinforcement learning

Multi-agent reinforcement learning investigates the problem where multiple agents learn and make decisions simultaneously in a shared environment. Each agent learns through interactions with both the environment and other agents, with the goal of optimizing their collective long-term returns. A MARL problem is formulated as a Markov game, represented as a tuple $\langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma \rangle$. Here, $\mathcal{N} = \{1, \ldots, n\}$ is a set of n agents, $\mathcal{S}$ is the state space, $\mathcal{A} = \prod_{i=1}^{n} \mathcal{A}^i$ is the product of action spaces of the n agents, known as the joint action space, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ represents the transition probability kernel, $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ denotes the reward function, and $\gamma \in [0,1)$ is the discount factor. Each agent i is associated with a parameter vector θ^i, which is combined with the parameter vectors of other agents to form a joint parameter vector $\boldsymbol{\theta}$.

At a time step $t \in \mathbb{N}$, an agent i takes an action $a_t^i \in \mathcal{A}^i$ drawn from its stochastic policy $\pi_\theta^i(\cdot|s_t)$ parametrized by θ^i. Similarly, other agents simultaneously give the joint action $\boldsymbol{a}_t = (a_t^1, \ldots, a_t^n) \in \mathcal{A}$, which drawn from the joint policy $\boldsymbol{\pi}_\theta(\boldsymbol{a}_t|s_t) = \prod_{i=1}^{n} \pi_{\boldsymbol{\theta}}^i(a_t^i|s_t)$. The total reward at time $t \in \mathbb{N}$ is defined as $R_t \triangleq \sum_{k=0}^{\infty} \gamma^k r_{t+k}$. The trajectory $\tau = \langle s_t, \boldsymbol{a}_t, r_t \rangle_{t=0}^{\infty}$ can be represented in the form of a sequence, which includes states, action, and rewards received by the agents in an interaction with the environment Then, the state value function $V_{\boldsymbol{\theta}}$ and the state-action value function $Q_{\boldsymbol{\theta}}$ are defined as (11) and (12)

$$V_{\boldsymbol{\theta}}(s) \triangleq \mathbb{E}_{a_{0:\infty} \sim \boldsymbol{\pi}_\theta, s_{1:\infty} \sim \mathcal{P}}[R_0 | s_0 = s], \tag{11}$$

$$Q_{\boldsymbol{\theta}}(s, \boldsymbol{a}) \triangleq \mathbb{E}_{s_{1:\infty} \sim \mathcal{P}, a_{0:\infty} \sim \boldsymbol{\pi}_\theta}[R_0 | s_0 = s, a_0 = \boldsymbol{a}]. \tag{12}$$

In a fully-cooperative game, the multi-agent policy gradient theorem provides the gradient of $\mathcal{J}(\boldsymbol{\theta})$ with respect to the parameter of the ith agent, which can be formulated as (13)

$$\nabla_{\theta^i} \mathcal{J}(\boldsymbol{\theta}) = \mathbb{E}_{s_{0:\infty} \sim d_\theta^{0:\infty}, a_{0:\infty}^{-i} \sim \boldsymbol{\pi}_\theta^{-i}, a_{0:\infty}^i \sim \pi_\theta^i} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\boldsymbol{\theta}}(s_t, \boldsymbol{a}_t^{-i}, a_t^i) \cdot \nabla_{\theta^i} \log \pi_{\boldsymbol{\theta}}^i(a_t^i|s_t) \right], \tag{13}$$

where $\mathscr{J}(\boldsymbol{\theta}) \triangleq \mathbb{E}_{s\sim d^0}[V_{\boldsymbol{\theta}}(s)]$ is the expected total reward, $-(i_1,\ldots,i_k)$ denotes the set of all agents excluding $i_1,\ldots,i_k$. Note that for $k=0$, $Q_{\boldsymbol{\theta}}(s_t, \boldsymbol{a}^{(i_1,\ldots,i_k)})$ is the state value function, and for $k=n$, this is the state-action value function.

3.2. OB-MAPPO algorithm

In heterogeneous multi-agent systems (e.g., UAVs and USVs), the conventional parameter-sharing mechanism in MAPPO faces limitations due to discrepancies in action spaces and difficulties in credit assignment. These issues can prevent the policy from converging to an optimal solution. Moreover, as the variance of gradient estimates grows rapidly with the number of agents, the performance of standard policy gradient methods degrades, resulting in biased action selection. To address these problems, an optimal baseline (OB) is proposed based on a heterogeneous advantage decomposition, which coordinates policy gradients to minimize variance, thereby improving parameter updates and enhancing training coordination among heterogeneous agents.

In any cooperative Markov game, given a joint policy $\boldsymbol{\pi}_\theta$, for any state s, and any agent subset $i_{1:m}$, the traditional advantage function can be established as (14)

$$\begin{aligned} A^{i_{1:m}}_{\boldsymbol{\pi}_\theta}(s, \boldsymbol{a}^{i_{1:m-1}}) &= Q^{i_{1:m}}_{\theta}(s, \boldsymbol{a}^{i_{1:m-1}}) - V_\theta(s) \\ &= \sum_{j=1}^{m}\left[Q^{i_{1:j}}_{\theta}(s, \boldsymbol{a}^{i_{1:j}}) - Q^{i_{1:j-1}}_{\theta}(s, \boldsymbol{a}^{i_{1:j-1}})\right] \\ &= \sum_{j=1}^{m} A^{i_j}_{\boldsymbol{\pi}_\theta}(s, \boldsymbol{a}^{i_{1:j-1}}, \boldsymbol{a}^{i_j}). \end{aligned} \tag{14}$$

Based on the principle of mirror learning (Kuba et al., 2022), for $i \in \mathscr{N}$, the heterogeneous agent drift functional mapping is defined as:

$$\mathscr{D}^i := \prod \times \prod \times \mathbb{P}(-i) \times \mathscr{S} \to \left\{ \mathscr{D}^i_{\boldsymbol{\pi}_\theta}(\cdot|s, \overline{\boldsymbol{\pi}}_{\boldsymbol{\theta}}^{j_{1:m}}) : \mathscr{P}(\mathscr{A}^i) \to \mathbb{R} \right\}, \tag{15}$$

where $\mathscr{D}^i_{\pi}(\pi^i|s, \overline{\pi}^{j_{1:m}}) \in \mathscr{P}(\mathscr{S})$ is a state distribution that can depend π on $\overline{\pi}$, which can be expressed in the form of expectations:

$$\mathscr{D}^v_{\boldsymbol{\pi}_\theta}(\overline{\boldsymbol{\pi}}_{\boldsymbol{\theta}}) = \mathbb{E}_{s\sim\nu^{\overline{\pi}}_{\pi}}\left[\mathscr{D}^i_{\boldsymbol{\pi}_{\boldsymbol{\theta}}}({\pi_{\boldsymbol{\theta}}}^i|s, \overline{\boldsymbol{\pi}}_{\boldsymbol{\theta}}^{j_{1:m}})\right], \tag{16}$$

where $\nu^{\overline{\pi}}_{\pi}$ ensures that expectations are consistent under both π and $\overline{\pi}$, with the differences between them being monotonically non-decreasing. This drift functional mapping can be interpreted as taking the conditional expectation over other agents' behaviours, thereby transforming the coupled joint advantage into decomposable local components. As a result, the uncertainty in joint policy gradient estimation is reduced, leading to lower variance and more stable convergence. For $i \in \mathscr{N}$ and $j^{1:m} \in \mathbb{P}(-i)$, the advantage function established based on drift functional can be given as (17)

$$\left[\mathscr{M}^{(\pi^i)}_{\mathscr{D}^i, \overline{\pi}^{j_{1:m}}} A_{\pi_{\boldsymbol{\theta}}}\right](s) = \mathbb{E}_{\boldsymbol{a}^{j_{1:m}}\overline{\pi}^{j_{1:m}}\sim\overline{\pi}^{j_{1:m}}, \boldsymbol{a}^i\sim\pi^i}\left[A^i_{\boldsymbol{\pi}_\theta}(s, \boldsymbol{a}^{j_{1:m}}, \boldsymbol{a}^i)\right] - \mathscr{D}^i_{\pi}(\pi^i_{\boldsymbol{\theta}}|s, \overline{\boldsymbol{\pi}}_{\boldsymbol{\theta}}^{j_{1:m}}). \tag{17}$$

This implies that sequential updates constitute an effective approach for identifying directions of performance improvement (i.e., joint actions for which the advantage function is positive). That is, suppose the agents act in an arbitrary order $i_{1:n}$. Let agent i_1 take action $a_t^{i_1}$ such that $A^{i_1}_{\boldsymbol{\pi}_\theta}(s, a_t^{i_1}) > 0$, and then, for the remaining $m = 2,\ldots,n$, each subsequent agent i_m take action $a_t^{i_m}$ such that $A^{i_m}_{\boldsymbol{\pi}_\theta}(s, a_t^{i_{1:m-1}}, a_t^{i_{1:m}}) > 0$. The joint advantage can be decomposed into sequential evaluations, thereby fully accounting for the influence of prior agent behaviour during the strategy update process. This function effectively marginalises other agents' action expectations and addresses the credit assignment problem in MARL through a complementary decomposition strategy.

For the MAPPO algorithm, which employs centralised training and decentralised execution (CTDE), incorporating a baseline $b(s_t, a^{-i})$ into the joint function $Q_{\boldsymbol{\theta}}(s, \boldsymbol{a})$ to maintain its unbiasedness can be expressed as:

$$\mathrm{g}^{\mathrm{i}}_{\mathrm{C}}(\mathrm{b}) = \sum_{\mathrm{t}=0}^{\infty}\gamma^{\mathrm{t}}\left[\mathrm{Q}_{\theta}(\mathrm{s}_{\mathrm{t}}, \mathrm{a}_{\mathrm{t}}^{-\mathrm{i}}, \mathrm{a}_{\mathrm{t}}^{\mathrm{i}}) - \mathrm{b}(\mathrm{s}_{\mathrm{t}}, \mathrm{a}_{\mathrm{t}}^{-\mathrm{i}})\right]\nabla_{\theta^{\mathrm{i}}}\log\pi^{\mathrm{i}}_{\theta}(\mathrm{a}^{\mathrm{i}}_{\mathrm{t}}|\mathrm{s}_{\mathrm{t}}). \tag{18}$$

On the basis of the heterogeneous advantage decomposition, the variance is further decomposed, and the variance of the joint advantage is expressed as the sum of the expectations of the conditional variances of the local advantages of individual agents, which can be formulated as:

$$\begin{aligned} \mathrm{Var}_{\mathrm{a}\sim\pi_\theta}\left[\mathscr{M}^{(\pi_\theta)}_{\mathscr{D}^{\mathrm{i}}, \overline{\pi}^{\mathrm{j}_{1:m}}} \mathrm{A}_{\pi_\theta}\right] = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathbb{E}_{\mathrm{a}^1\sim\pi^1_{\theta},\ldots,\mathrm{a}^{\mathrm{i}-1}\sim\pi^{\mathrm{i}-1}_{\theta}} \\ \left[\mathrm{Var}_{\mathrm{a}^{\mathrm{i}}\sim\pi^{\mathrm{i}}_{\theta}}\left[\mathscr{M}^{(\pi_\theta)}_{\mathscr{D}^{\mathrm{i}}, \overline{\pi}^{\mathrm{j}_{1:m}}} \mathrm{A}_{\pi_\theta}(\mathrm{s}, \mathrm{a}^{(1,\ldots,\mathrm{i}-1)}, \mathrm{a}^{\mathrm{i}})\right]\right], \end{aligned} \tag{19}$$

where $\mathbf{Var}_{a\sim\pi_\theta}[*]$ represents the variance of the advantage function of the agent when taking action a under the joint strategy π_θ. Note that the variance of the joint advantage does not exceed the sum of the variances of the local advantages of all agents. Based on this, we decompose the single-step variance of the CTDE estimator into three components:

$$\begin{aligned} \mathbf{Var}\left[g^i_C(b)\right] = \mathbf{Var}_{s_t}\left[\mathbb{E}_a\left[g^i_C(b)\right]\right] + \mathbb{E}_{s_t}\left[\mathbf{Var}_{a^{-i}}\left[\mathbb{E}_{a^i}\left[g^i_{C,t}(b)\right]\right]\right] \\ + \mathbb{E}_{s_t,a^{-i}}\left[\mathbf{Var}_{s_t,a^{-i}}\left[g^i_{C,t}(b)\right]\right]. \end{aligned} \tag{20}$$

In Equation (20), the first and second terms on the right-hand side correspond to the state variance and the variance related to other agents (independent of b), respectively, while the third term is the variance associated with the agent's own action (dependent on b). Therefore, minimising the total variance is equivalent to minimising:

$$\mathbb{E}_{s,a^{-i}}\left[\mathbf{Var}_{a^i\sim\pi^i_\theta} Q_{\boldsymbol{\theta}}(s_t, a_t^{-i}, a_t^i) - b(s_t, a_t^{-i})\right]\nabla_{\theta^i}\log\pi^i_{\boldsymbol{\theta}}(a^i_t|s_t)\Big]. \tag{21}$$

For Formula (21), after expanding the variance, taking the derivative with respect to $b(s_t, a_t^{-i})$, and setting the derivative equal to zero, it is easy to obtain:

$$b_{op}(s, a^{-i}) = \frac{\mathbb{E}_{a^i,\pi^i_\theta}\left[Q_{\boldsymbol{\theta}}(s_t, a_t^{-i}, a_t^i)\cdot\left\|\nabla_{\theta^i}\log\pi^i_{\boldsymbol{\theta}}(a^i_t|s_t)\right\|^2\right]}{\mathbb{E}_{a^i,\pi^i_\theta}\left[\left\|\nabla_{\theta^i}\log\pi^i_{\boldsymbol{\theta}}(a^i_t|s_t)\right\|^2\right]}. \tag{22}$$

For an arbitrary policy $\pi^i_{\boldsymbol{\theta}}(a^i_t|s_t) = \pi^i(a^i_t|\psi^i_\theta(s))$, where $\psi^i_\theta(s)$ is the output layer of the policy network and $\nabla_{\psi^i_\theta(s)}\log\pi^i$ has a dimensionality equal to that of the action space, the computational cost remains low (substantiated in Section 4.3). By applying the chain rule to decompose the gradient, a neural network approximation of b_{op} can be obtained:

$$b^*(s, a^{-i}) = \frac{\mathbb{E}_{a^i,\pi^i_\theta}\left[Q_{\boldsymbol{\theta}}(s_t, a_t^{-i}, a_t^i)\cdot\left\|\nabla_{\psi^i_\theta(s)}\log\pi^i(a^i_t|\psi^i_\theta(s))\right\|^2\right]}{\mathbb{E}_{a^i,\pi^i_\theta}\left[\left\|\nabla_{\psi^i_\theta(s)}\log\pi^i(a^i_t|\psi^i_\theta(s))\right\|^2\right]}. \tag{23}$$

The OB refined through the advantage decomposition can be integrated into $Q_{\boldsymbol{\theta}}$ to optimize gradient variance, which can be expressed as:

$$\widehat{\chi}^i(s, a^{-i}, a^i) = \widehat{Q}_{\boldsymbol{\theta}}(s, a^{-i}, a^i) - b^*(s, a^{-i}). \tag{24}$$

Based on the aforementioned improvements, the objective function of the OB-MAPPO algorithm can be expressed as:

$$\begin{aligned} \mathscr{J}(\boldsymbol{\theta}) = \sum_{i=1}^{n}\mathbb{E}_{s\sim\rho_{\theta_{old}}, \boldsymbol{a}\sim\pi_{\theta_{old}}}\left[\min\left(\frac{\pi_\theta(a^i|s)}{\pi_{\theta_{old}}(a^i|s)}\right)\cdot\widehat{\chi}^i(s, \boldsymbol{a})\right. \\ \left., \mathrm{clip}\left(\frac{\pi_\theta(a^i|s)}{\pi_{\theta_{old}}(a^i|s)}, 1-\varepsilon, 1+\varepsilon\right)\cdot\widehat{\chi}^i(s, \boldsymbol{a})\right]. \end{aligned} \tag{25}$$

Remark 1. Coordination is achieved at the optimisation level via the OB mechanism. By decomposing the joint advantage into conditional local advantages, each agent updates its policy under the expectation of other agents' behaviours. This enables implicit coordination without explicitly modelling the full joint action space. Furthermore, the OB

formulation reduces the effective optimisation complexity. Instead of directly operating in an exponentially growing joint action space, the learning process is transformed into a series of lower-dimensional updates. The analytically derived baseline also reduces gradient variance, improving training stability. Overall, this decoupled representation and coupled optimisation paradigm accommodates heterogeneity while mitigating the curse of dimensionality.

3.3. HCCAD strategy under OB-MAPPO framework

Within the OB-MAPPO framework, UAV and USV are considered heterogeneous agents interacting with a dynamic environment, with this process schematically represented in Fig. 5. The implementation of the HCCAD strategy necessitated the following designs to the OB-MAPPO framework.

1) State space design: The fusion of sequential visual data from the UAV camera and LiDAR observations from the USV facilitates the construction of a comprehensive state space. This integration diminishes the reliance on global information, thereby augmenting the generalization capacity of the HCCAD strategy. The state space is established as follows:

$$\mathscr{S}^a = \left\{ s_t^a := \left(\boldsymbol{p}_{im}^a, \boldsymbol{p}^a, v_t^a, f_t^{cam} \right) \right\}, \tag{26}$$

$$\mathscr{S}^s = \left\{ s_t^s := \left(\boldsymbol{p}^s, o_t^{LiDAR} \right) \right\}, \tag{27}$$

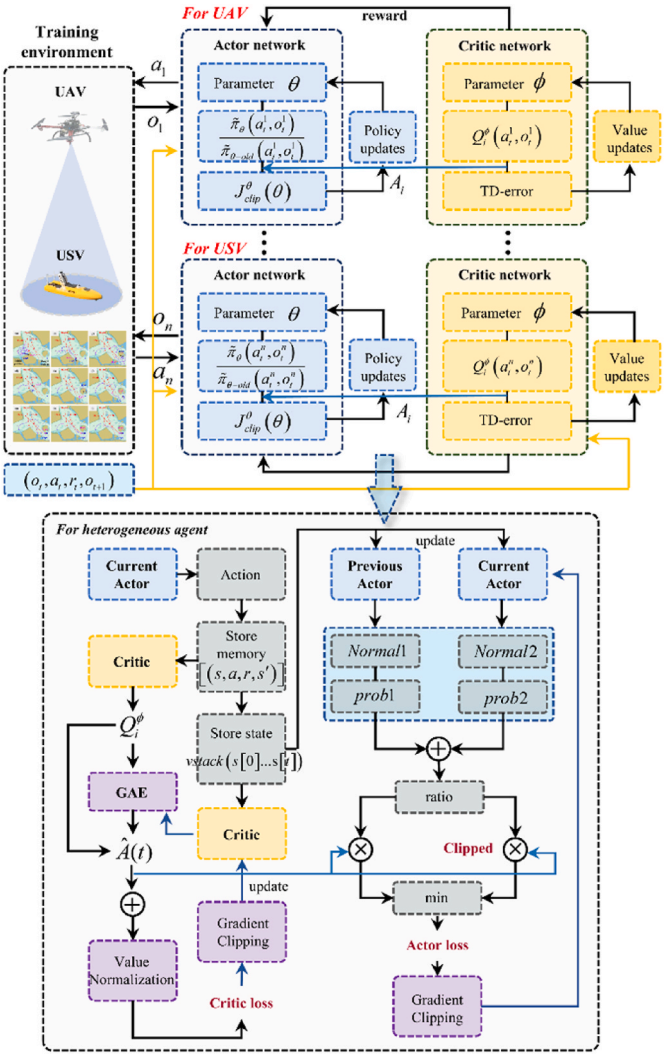


Fig. 5. The interaction process between the OB-MAPPO and environment.

$$\mathscr{S} := \left\{ \mathscr{S}^a, \mathscr{S}^s, \boldsymbol{g}_g \right\}, \tag{28}$$

where $\mathscr{S}^a$ and $\mathscr{S}^s$ are the state space of the UAV and USV, respectively, $\boldsymbol{g}_g$ is the target position, $\boldsymbol{p}_{im}^a(x_{im}, y_{im})$ is the projection position of the UAV, $\boldsymbol{p}^a$ is the current position of the UAV, v_t^a is the current speed of the UAV, $f_{ti}^{cam} \in \mathbb{R}^{1\times n_{cam}}$ represents the real-time camera observed image data of the UAV, and n_{cam} indicates the number of samples taken by the camera. $\boldsymbol{p}^s$ is the current position of the USV, $o_t^{LiDAR} \in \mathbb{R}^{1\times n_{lm}}$ represents the real-time LiDAR observed data of USV, and n_{lm}, $m = 1, \ldots, N$ is the number of set of angular-distance pairs in 360° field of view by the LiDAR.

2) Action space design: The definition of the action space fundamentally governs the scope of the decision-making landscape. It is a critical determinant of an agent's learning potential, efficiency, and the final performance of the coordinated strategy. A high-dimensional, continuous action space is therefore adopted for the HCCAD policy to achieve superior control fidelity and facilitate complex, smooth motion trajectories. Given the differing kinematic properties of UAV and USV, we define their action spaces independently:

$$\mathscr{A}^a = \left\{ a_t^a := \left[\left(\delta_a^r \sim \delta_a^l \right), \left(v_{\min}^a \sim v_{\max}^a \right) \right] \right\}, \tag{29}$$

$$\mathscr{A}^s = \left\{ a_t^s := \left[\left(\delta_s^{port} \sim \delta_s^{starb} \right) \right] \right\}, \tag{30}$$

where $\mathscr{A}^a$ and $\mathscr{A}^s$ are the action space of the UAV and USV, δ_a^r and δ_a^l respectively is the maximum steering angles for UAV to turn left and right. $v_{\min}^a$ and $v_{\max}^a$ are the minimum and maximum speed of the UAV, δ_s^{port} and δ_s^{starb} indicate the maximum of port and starboard rudder angle for the USV.

3) Network structure design: The OB-MAPPO is built upon the actor-critic training framework, in which the actor network generates actions for interacting with the environment, while the critic network assesses the quality of these actions. Given that the state space incorporates observations from different sensors, we employ a multi-modal fusion network to process the cross-modal information (e.g., images, radar observation vectors, and coordinates). The architecture of this network is illustrated in Fig. 6.

Specifically, the multimodal feature fusion network comprises three branches: a primary branch (PB), an auxiliary branch (AB), and a reconciliation branch (RB). The PB and AB alternately focus on LiDAR observation vectors and image features, thereby facilitating effective cross-modal interaction. The RB centres on a shared reconstruction task, encouraging the network to learn general feature representations, harmonize optimisation directions, and mitigate overfitting.

After acquiring the shared features, both the PB and AB employ a cross-fusion gated mechanism (CFGM) to further extract specific features corresponding to different fusion types, thereby adaptively stabilizing cross-task interactions. Owing to its well-established capability for robust global feature extraction (Cheng et al., 2025), we implement the CFGM using efficient Swin Transformer blocks.

Compared to the traditional Transformer, the Swin Transformer replaces the multi-head self-attention (MSA) with a combination of window-based MSA (W-MSA) and shifted window MSA (SW-MSA). By partitioning the image into non-overlapping windows in a uniform manner, this design significantly reduces computational complexity, which can be expressed as:

$$\begin{cases} \hat{z}^l = \mathrm{W-MSA}\left(\mathrm{LayerNorm}\left(z^{l-1}\right)\right) + z^{l-1} \\ z^l = \mathrm{MLP}\left(\mathrm{LayerNorm}\left(\hat{z}^l\right)\right) + \hat{z}^l \\ \hat{z}^{l+1} = \mathrm{SW-MSA}\left(\mathrm{LayerNorm}\left(\hat{z}^l\right)\right) + z^l \\ z^{l+1} = \mathrm{MLP}\left(\mathrm{LayerNorm}\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1}, \end{cases} \tag{31}$$

where $\hat{z}^l$ and z^l respectively represent the output features of the (S)W-MSA module and the output features of the multi-layer perceptron

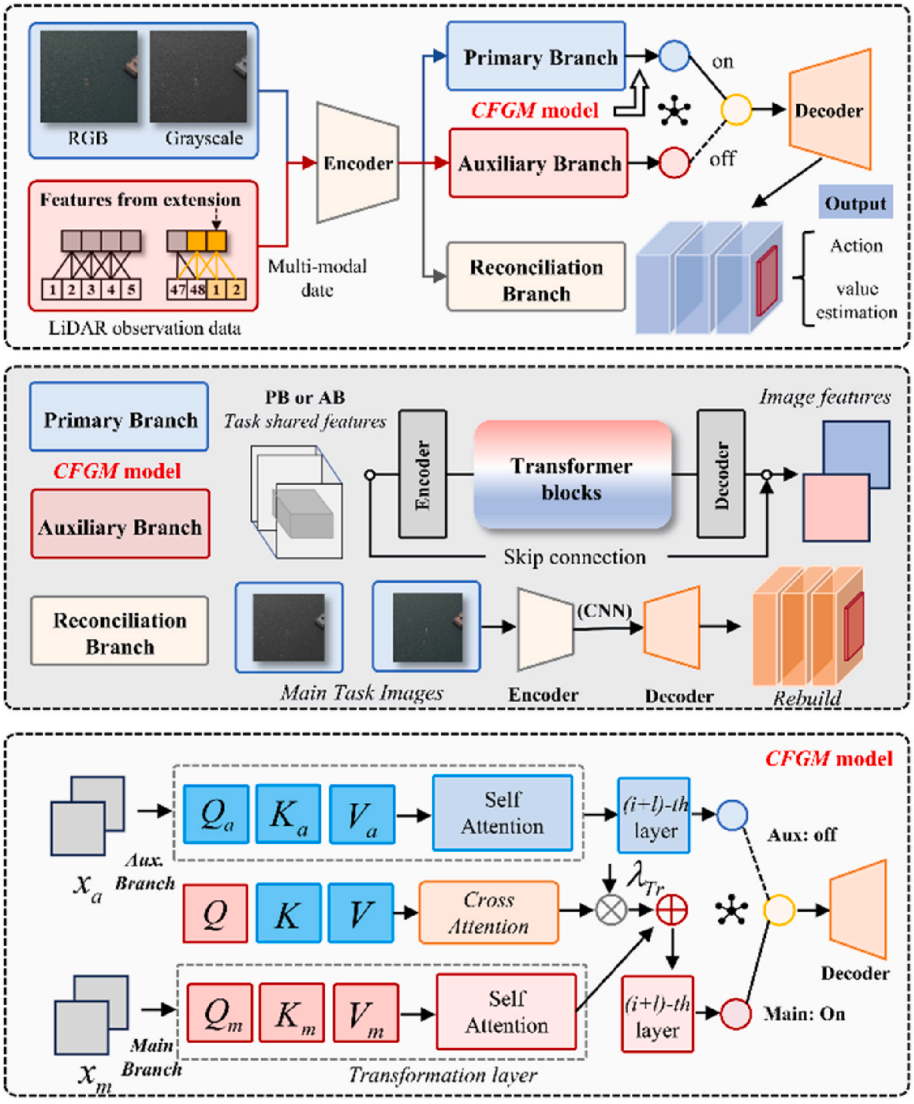


Fig. 6. The multi-modal feature fusion neural network architecture based on Swin Transformer.

(MLP) module of the *l*th block. Based on the above, the computational process of the Swin Transformer can be expressed as:

$$\text{Attention}(Q_{ST}, K_{ST}, V_{ST}) = \text{Softmax}\left(\frac{Q_{ST}{K_{ST}}^{T}}{\sqrt{d_k}} + \mathbf{B}_{ST}\right)\cdot V_{ST}, \tag{32}$$

where Q_{ST} , K_{ST}, and V_{ST} represent query, key, and value matrix, $d_k = C_{ST}/h$, C_{ST} is the feature channel embedding dimension, h is the number of feature head, and $\mathbf{B}_{ST}$ is the relative position offset matrix, which encodes the complete set of relative positional relationships between any two pixels in a window, thereby introducing a bias term into the attention mechanism that significantly improves the model's geometric understanding. Then, the cross-attention fusion mechanism applied to CMGF can be formulated as:

$$\begin{cases} \widehat{x}_{PB} = \text{Attention}(x_{PB}) \\ x_{PAB} = \widehat{x}_{PB} + \lambda\cdot\text{CrossAttention}(\widehat{x}_{PB}, x_{AB}), \end{cases} \tag{33}$$

where x_{PB} and x_{AB} are feature representations of PB and AB, and λ denote the learnable parameters that modulate the influence intensity of AB on PB, effectively preventing auxiliary information overload. In the end, the network loss of multimodal feature fusion can be expressed as:

$$\begin{cases} \mathscr{L}_t = \mathscr{L}_{RB} + \mathscr{L}_{PB} + \mathscr{L}_{AB} \\ \mathscr{L}_{RB} = \mathscr{L}_{ssim}(I_{RB}, I_{LiDAR}) + \mathscr{L}_{mse}(I_{RB}, I_{LiDAR}) \\ \mathscr{L}_{PB} = w_{im}\cdot\mathscr{L}_{mse}(I_f, I_{im}) + w_{LiDAR}\cdot\mathscr{L}_{mse}(I_f, I_{LiDAR}) \\ \mathscr{L}_{AB} = (I_f, I_{LiDAR}), \end{cases} \tag{34}$$

where $\mathscr{L}_{RB}$, $\mathscr{L}_{PB}$, and $\mathscr{L}_{AB}$ are the RB, PB, and AB network loss, respectively, I_{RB} represents the output vector of the RB, I_{im}, and I_{LiDAR} are the extracted feature vector by images and LiDAR data, $\mathscr{L}_{mse}$ represents the mean square error loss, w_{im} and w_{LiDAR} are the adaptive weights derived from the fusion of image and LiDAR loss components.

4) Reward function design: The design of the reward function should ensure that the UAV-USV not only maximizes cumulative rewards but also, through multiple iterations of neural network training, achieves heterogeneous collaboration and avoids TSs in unknown environments. The proposed reward function is designed as follows:

$$\begin{cases} F_R = \lambda_R {R_R}^{\mathrm{T}} \\ R_R = \left[R_g, R_{ce}^a, R_c^s, R_i^s\right] \\ \lambda_R = \left[\lambda_g, \lambda_{ce}^a, \lambda_c^s, \lambda_i^s\right], \end{cases} \tag{35}$$

where λ_R is the function weights, R_g is the guidance reward, R_{ce}^a is the

collaborative reward, R_c^s and R_i^s are the collision avoidance reward and immediate danger reward, respectively.

The guidance reward serves to direct the UAV-USV to the target location. A collision penalty is further introduced to ensure the avoidance of TSs and obstacles, which can be formulated as follows:

$$R_g = \begin{cases} -k_1, & \text{ifcollision}, \\ -\omega_g \cdot \ln\left\| \left(\boldsymbol{g}_g - \boldsymbol{p}^s \right) + 1 \right\|, & \text{otherwise}, \end{cases} \tag{36}$$

where $\boldsymbol{g}_g$ is the target position, $\boldsymbol{p}^s$ is the current position of the USV, and ω_g is the weight parameter.

The collaboration reward ensures synergistic cooperation between the UAV and USV. This is achieved by defining the UAV's desired tracking position as the projection of the USV's 2D surface coordinates onto the 3D aerial space. The collaboration reward then penalises the misalignment error, incentivising the UAV to adjust its trajectory for precise positioning, which is given by:

$$R_{ce}^a = -\omega_{ce1} \left\| \boldsymbol{p}^a - \boldsymbol{p}^{ap} \right\|^2 - \omega_{ce2} \left\| \boldsymbol{p}_{im}^a - \boldsymbol{p}_{im}^{ap} \right\|^2, \tag{37}$$

where $\boldsymbol{p}^a$ and $\boldsymbol{p}^{ap}$ are current and predicted position of the UAV, $\boldsymbol{p}_{im}^a$ and $\boldsymbol{p}_{im}^{ap}$ are current and predicted projection position of the UAV, ω_{ce1} and ω_{ce2} are weight parameters.

To ensure that the trained HCCAD strategy produces manoeuvres during collision avoidance that comply with COLREGs, we incorporate the 13th to 15th of COLREGs into the design of the collision avoidance reward. Specifically, when TS intrudes into the USV's safety range, the USV receives a positive reward if its avoidance manoeuvre adheres to COLREGs, and a negative reward otherwise, which can be expressed as:

$$R_c^s = \begin{cases} \omega_{c1} \cdot (\exp|d_T| - k_2), & d_T > R_S, \\ -\omega_{c2} \cdot |R_S - d_T|, & (d_T < R_S) \cap \varpi_1, \\ k_3, & (d_T < R_S) \cap \varpi_2, \end{cases} \tag{38}$$

where d_T is the distance between USV and TS, R_S is the safety distance, ω_{c1} and ω_{c2} are weight parameters, ϖ_1 and ϖ_2 represent non-compliant and compliant USV manoeuvres with respect to COLREGs, respectively.

When a TS intrudes into the ship domain of USV or when the USV encounters a multi-ship encounter situation, it implies that the USV cannot safely avoid collisions on its own. To address this, we design an immediate danger reward, which can be expressed as:

$$R_i^s = \begin{cases} \omega_i \cdot (\omega_{c2} \cdot |R_S - d_T|), & (d_T < R_S) \cap \varpi_3, \\ 0, & \text{else}, \end{cases} \tag{39}$$

where ω_i is the weighting parameter, while ϖ_3 represents the manoeuvre of the USV deviating from the COLREGs when a TS intrudes into the USV's domain. Fig. 7 summarizes the specific process of solving HCCAD problem based on OB-MAPPO algorithm.

Remark 2. In this study, the reward function is formulated under a task-driven and safety-constrained paradigm, integrating a primary guidance term with auxiliary components for collaboration, collision avoidance, and immediate danger. This design alleviates reward sparsity while aligning policy optimisation with both task objectives and safety requirements. During early training, frequent collisions cause the agent to prioritise guidance and safety penalties, with limited influence from auxiliary terms. As learning progresses and experience accumulates, the agent increasingly exploits the coupling between collaboration accuracy and task success. Consequently, auxiliary rewards facilitate more efficient exploration of the state-action space and promote coordinated behaviours.

4. Experiment

4.1. Configuration of simulation

To validate the effectiveness of the proposed OB-MAPPO algorithm in addressing the HCCAD problem between UAV and USV, multiple simulation scenarios are designed. The experimental platform is configured with an Intel Core i9-11900H CPU, NVIDIA RTX 3090 GPU, Gazebo 11.14, and PyTorch 1.13.0, and runs on Ubuntu 20.04 with ROS Noetic. Furthermore, the USV and UAV in the simulations are modelled after the Virtual Robot-X (VRX) and PX4 physical platforms (Bingham et al., 2019), respectively. The hydrodynamic derivatives and model parameters of the simulated USV and UAV are provided in references (Cui et al., 2023a; Wang et al., 2023a). The hyperparameters of the OB-MAPPO algorithm are illustrated in Table 1.

Algorithm 1: OB-MAPPO for solving the HACCAD problem

Initialize: Actor parameter θ, Critic parameter ϕ, the batch size N_b, the episodes number N_e, the empty replay buffer D_r, and the clipping parameter ε.
Set target parameters equal to main parameters $\theta_T \leftarrow \theta_\pi$, $\phi_T \leftarrow \phi$.
Repeat
  Select action $a_i^t \sim \pi_\theta(a_i^t \mid s_i^t)$, and observe state s with (26).
  Execute joint $\mathbf{a}^t = \{a_1^t, \ldots, a_N^t\}$ with (29) in the training environment.
  Observe next state s', reward r with (35), and done signal d to indicate whether s' is terminal.
  Store (s, a, r, s') in replay buffer D_r.
  If it's time to update **then**
    For each agent i
      Calculate drift advantage function with (17).
      Calculate TD error δ_t
      Calculate optimal baseline with (23)
      Calculate OB refined drift advantage function with (24)
    End for
    Calculate GAE with Actor and Critic parameters
    For each gradient-updating step in the Actor network **do**
      Update actor loss of gradient ascent with (25).
      Update the Actor network θ.
    End for
    For each gradient-updating step in the Critic network **do**
      Update critic loss of gradient descent with:
      $L_{critic} = E[\max((V_\phi(s_t) - R_t)^2, (\text{clip}(V_\phi(s_t), V_{old}, \varepsilon) - R_t)^2)]$.
      Update the Critic network ϕ.
    End for
    Empty replay buffer.
  End if
Until convergence

Fig. 7. The specific process of solving HCCAD problem based on OB-MAPPO algorithm.

Table 1
Hyper-parameters of the ob-mappo algorithm.

Parameter	value	Parameter	value
Discount factor	0.99	GAE parameter	0.95
clipping coefficient	0.2	Actor learning rate	3e-4
Critic learning rate	1e-3	Optimiser	Adam
Entropy coefficient	0.01	Value loss coefficient	0.5
Gradient clipping	0.5	Update epochs	10
Batch size	128	Replay buffer size	1e+5
Transformer embedding	128	attention head numbers	4
Swin Transformer layers	2	Training episodes	5e+4

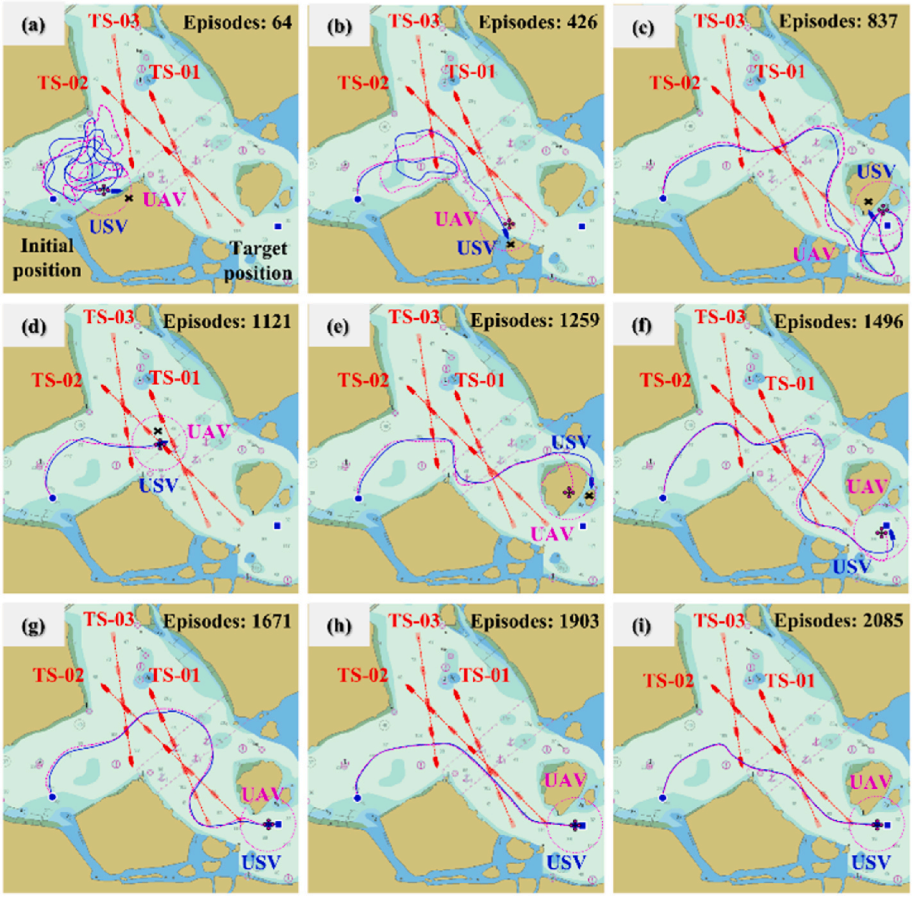


Fig. 8. The HCCAD model training environment and training process.

4.2. Configuration of simulation

The training environment for the HCCAD strategy, as illustrated in Fig. 8, incorporates essential elements encountered during navigation, including three TSs, narrow waters, open waters, and static obstacles. In the early stage of training (Fig. 8a), the trajectories of the UAV and USV exhibit prominent exploratory characteristics. In the complex water environment with obstacles, the agents extensively explore the state space and action mappings, resulting in relatively disordered trajectories. This indicates that the HCCAD model remains in the strategy exploration phase, with preliminary attempts to learn the guidance reward.

As the number of training episodes increases (Fig. 8b, c, d, e), the trajectories of the UAV and USV gradually demonstrate adaptability to environmental constraints and cooperative requirements. The agents begin to form more purposeful movement trends, and preliminary collision avoidance behaviours for obstacles emerge. In the mid-to-late stage of training (Fig. 8f, g, h), the exploratory nature of the trajectories further decreases, while the exploitative performance significantly improves. The movement trajectories of the UAV and USV become smoother and more efficient, and the collision avoidance actions for obstacles become more predictive. The temporal and spatial synchronisation of cooperative behaviours is enhanced, indicating that, through numerous interactive learning episodes, the model has constructed a more robust collision-avoidance and cooperative strategy, and that strategy convergence gradually emerges.

Eventually, in the late stage of training (Fig. 8i), the trajectories of the UAV and USV achieve highly optimised collaboration collision avoidance. Their movement trajectories not only meet the safety requirements for avoiding multiple obstacles in complex waters but also form an efficient task path through collaboration decision-making (during the process from the initial position to the target position, the trajectory is smooth without redundant exploration).

4.3. Training comparison experiment

To comprehensively evaluate the performance of the proposed OB-

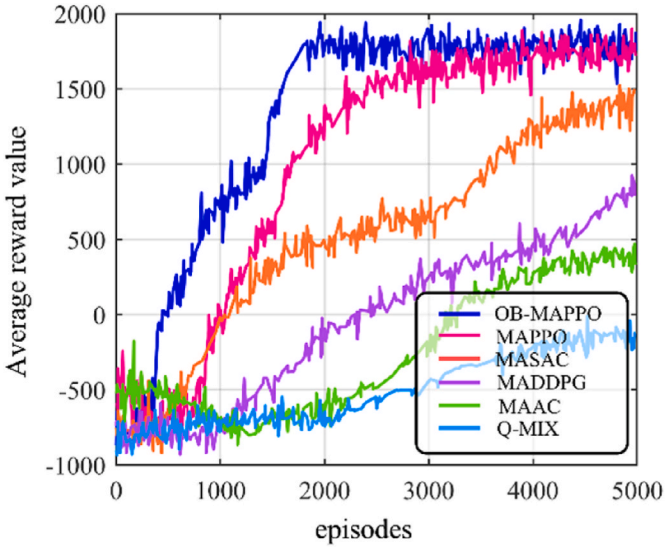


Fig. 9. Comparison of average reward value curve based on OB-MAPPO, MAPPO, MASAC, MADDPG, and Q-MIX algorithms.

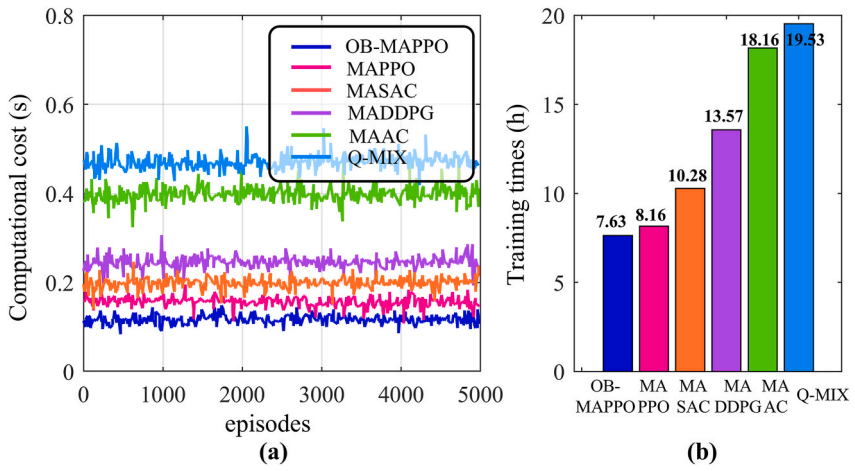


Fig. 10. Comparison of computational cost and training time based on OB-MAPPO, MAPPO, MASAC, MADDPG, and Q-MIX algorithms.

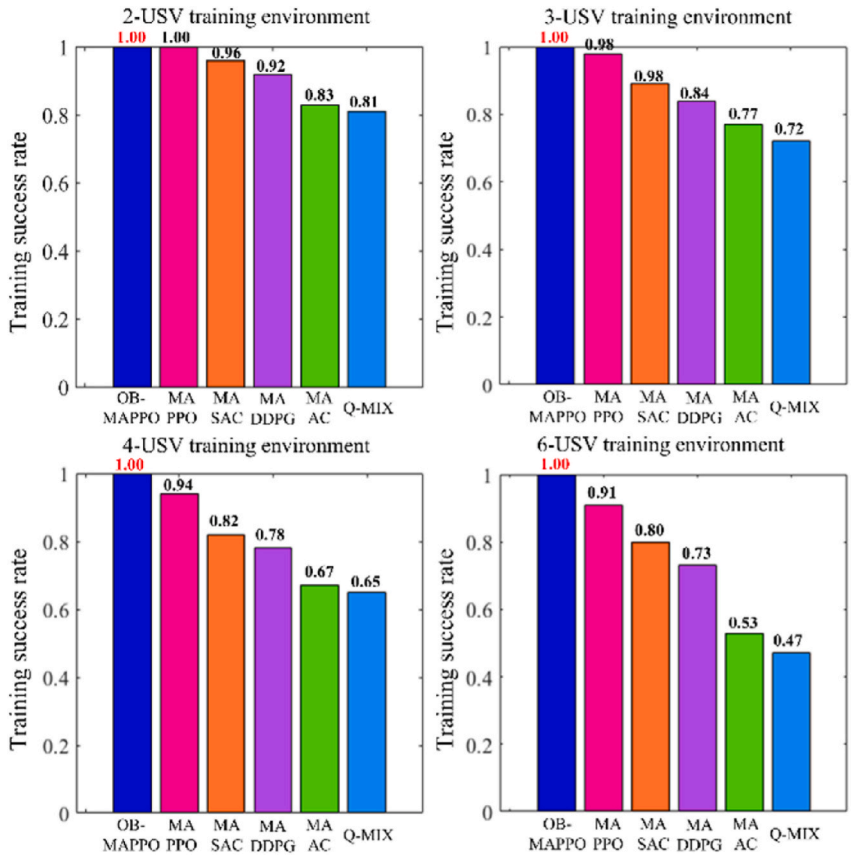


Fig. 11. Comparison of training success rate based on OB-MAPPO, MAPPO, MASAC, MADDPG, and Q-MIX algorithms in 2-USV, 3-USV, 4-USV, and 6-USV environment.

MAPPO algorithm against state-of-the-art MARL methods (i.e., MAPPO, MASAC, MADDPG, MAAC, and Q-MIX), four key metrics are analysed: average reward value during training, computational cost (CC), training time (TT), and training success rate (TSR) under different numbers of USVs. From the comparative data shown in Figs. 9–11, the following observations can be made.

(1) The OB-MAPPO exhibits the fastest convergence speed and achieves the highest ultimate average reward (Fig. 9.). From 1500 to 2000 episodes, the OB-MAPPO's reward curve rises sharply and stabilises at a high level, outperforming the MAPPO. In contrast, other algorithms (MASAC, MADDPG, MAAC, Q-MIX) demonstrate significantly slower convergence and lower terminal rewards.

(2) The OB-MAPPO maintains the lowest CC throughout all episodes, with minimal fluctuations (Fig. 10a). The MAPPO and MASAC also exhibit relatively low computational overhead, while the MADDPG, MAAC, and especially Q-MIX incur substantially higher costs. Lower CC implies OB-MAPPO can deliver superior performance with reduced hardware resource consumption. The TT result (Fig. 10b) aligns with the CC analysis, confirming that

OB-MAPPO not only reduces per-episode computation but also shortens overall training duration.

(3) An increase in the number of TSs leads to a decrease in the TSR across all algorithms (Fig. 11), which is consistent with the expectation that a denser environment poses a greater challenge. The OB-MAPPO achieves a near-perfect success rate (1.00) in all scenarios; even in large-scale settings (e.g., 6-USV environment), it outperforms other algorithms by a wide margin.

(4) The monotonicity constraint forces Q-MIX to assume that the joint action-value function is a monotonic combination of individual value functions. In multi-USV scenarios, interactions between agents are often non-monotonic (e.g., an agent's optimal action depends on non-additive contributions from others). The Q-MIX cannot model such dependencies, leading to suboptimal joint policies as the number of USVs increases.

(5) The MASAC and MADDPG achieve superior performance over MAAC by integrating a maximum entropy objective and double Q-networks, respectively, to alleviate value overestimation. In contrast, the OB-MAPPO outperforms MAPPO by combining the OB mechanism with heterogeneous advantage decomposition. This integration reduces variance, enabling more efficient parameter updates and enhanced training effectiveness.

4.4. Simulation verification

Through simulation experiments, the effectiveness of the HCCAD strategy trained using the OB-MAPPO algorithm is validated. The experiments include two-ship collision avoidance in classical encounter scenarios and multi-ship collision avoidance in complex environments. Furthermore, the environments in simulation are all real water areas in electronic navigational chart and the interference are set as follows: 35° for wave and wind direction; 0.15m for significant wave height; 4.0s for wave period; 0.30 m/s for mean wind speed.

For two-ship collision avoidance, four typical encounter scenarios are investigated: head-on, overtaking, port-crossing, and starboard-crossing. The simulation results are summarised in Table 2. The head-on encounter process is illustrated in Fig. 12. When the TS-01 intruded into the safety range of the USV, the USV executed a substantial starboard alteration to avoid collision. After the two ships had safely passed each other and all risk of collision was confirmed to be resolved, the USV made a slight port adjustment to return to its intended course. The overtaking encounter process is illustrated in Fig. 13. The TS-01, acting as the stand-on ship, maintains her course and speed, while the USV initiates a starboard alteration to safely complete the overtaking situation. In addition, the navigation trajectories for the starboard-crossing and port-crossing scenarios are illustrated in Figs. 14 and 15, respectively. In Fig. 14, the USV operates as the give-way ship. Once TS-01 entered its safety range, the USV initiated a pronounced starboard alteration to safely navigate the encounter. In Fig. 15, the TS-01, as the give way ship, failed to take appropriate collision avoidance actions in compliance with the COLREGs. Faced with an immediate danger situation, the USV deviated from the COLREGs by executing a substantial starboard alteration to negotiate the encounter.

Fig. 16 illustrates the collaboration collision avoidance process in a multi-ship encounter scenario. Initially, the USV and TS-03 were in a head-on situation, prompting the USV to alter her course to starboard to complete the encounter. To avoid TS-02 approaching from the starboard side, the USV executed an emergency alteration to port to ensure navigational safety. Subsequently, during encounters with TS-01 and TS-04,

Table 2
Simulation data of the two-ship encounter collision avoidance.

Encounter Type	Encounter angle (deg)	Minimum distance (m)	Collaboration error (m)	Navigation time (s)
Head-on	3.6	48.6	0.139	95.4
Overtaking	174.3	27.1	0.128	98.2
Starboard-crossing	73.8	44.7	0.134	101.9
Port-crossing	320.4	38.7	0.133	109.8

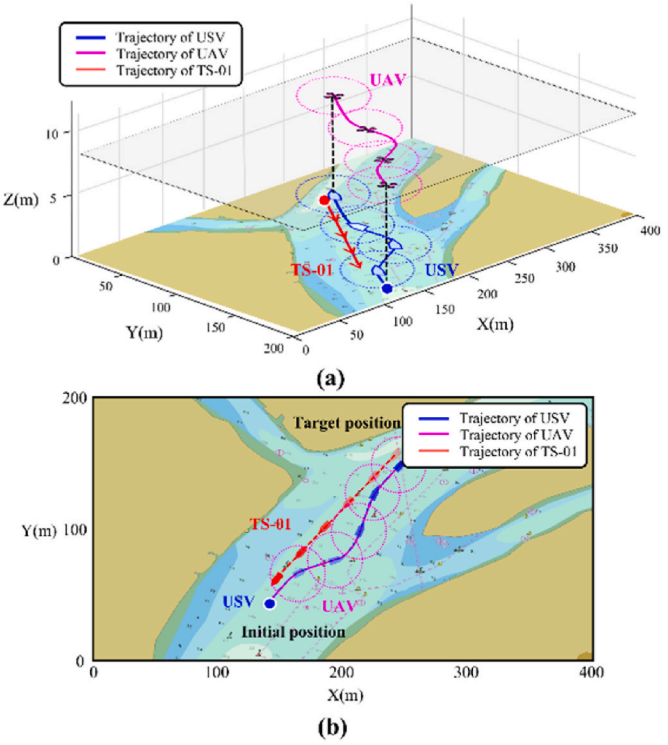


Fig. 12. The collision avoidance trajectories of UAV-USV in the head-on situation (a) 3D spatial view (b) 2D vertical view.

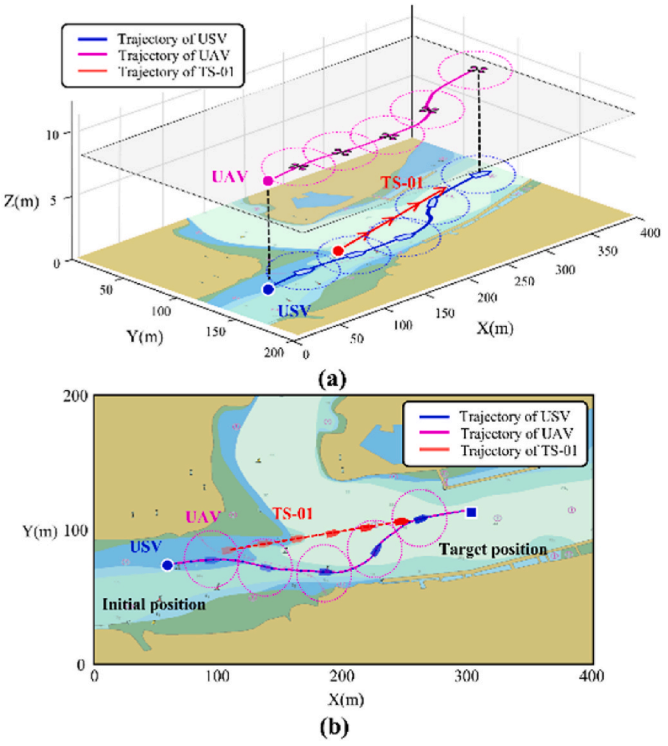


Fig. 13. The collision avoidance trajectories of UAV-USV in the overtaking situation (a) 3D spatial view (b) 2D vertical view.

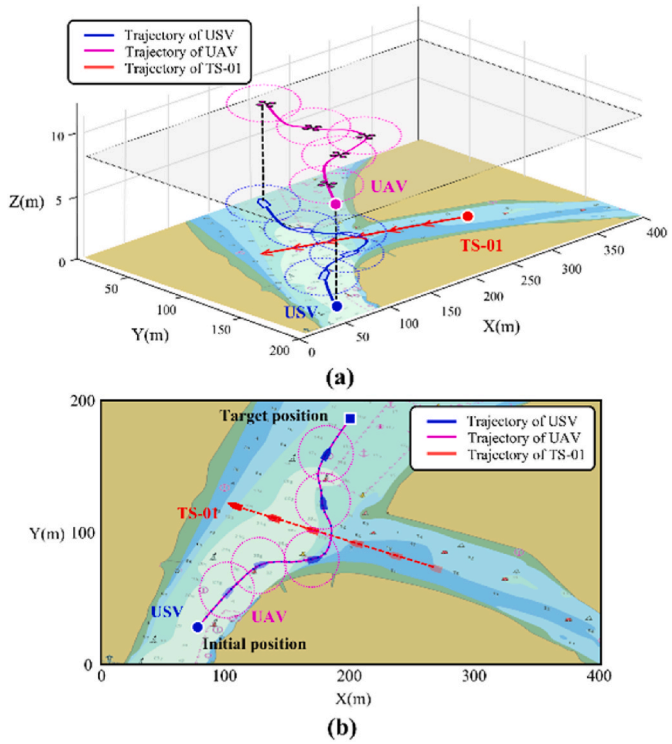


Fig. 14. The collision avoidance trajectories of UAV-USV in the starboard-crossing situation (a) 3D spatial view (b) 2D vertical view.

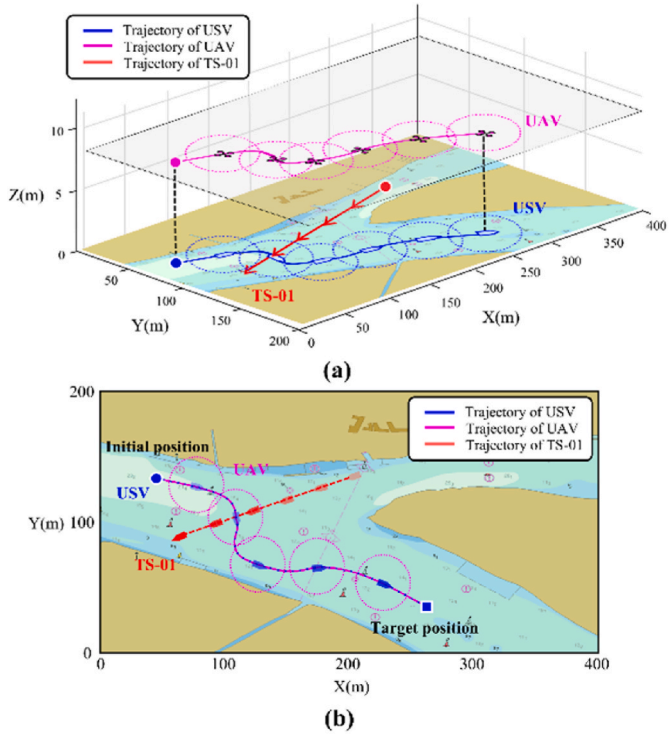


Fig. 15. The collision avoidance trajectories of UAV-USV in the port-crossing situation (a) 3D spatial view (b) 2D vertical view.

both ships entered the USV's ship domain. In response, the USV deviated from COLREGs by making a slight starboard alteration followed by an urgent port alteration, ultimately arriving safely at the target location. The time-varying curves of the USV's rudder angle, the UAV's steering

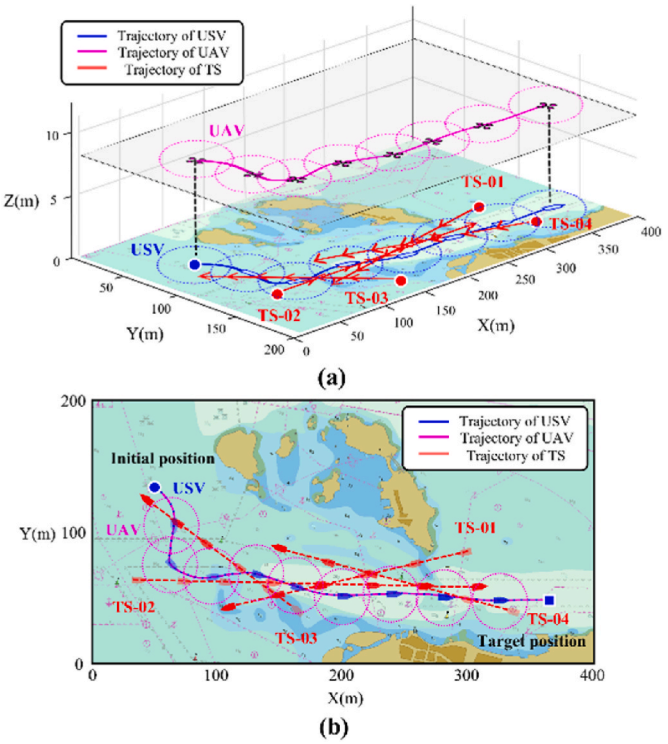


Fig. 16. The collision avoidance trajectory of UAV-USV in the situation of multiple ships encounter situation.

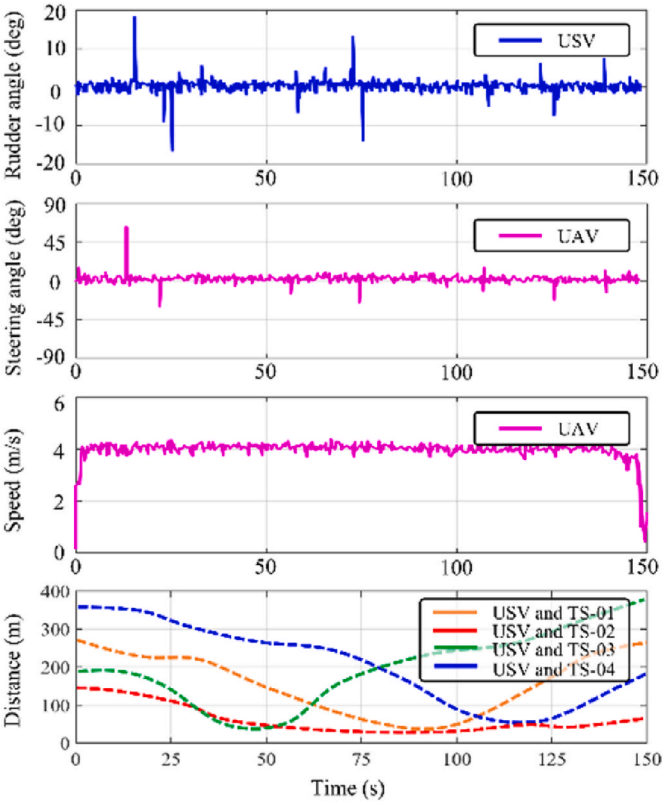


Fig. 17. The time-varying curves of the USV's rudder angle, the UAV's steering angle and speed, and the distance between the two ships during multiple ships encounter situation.

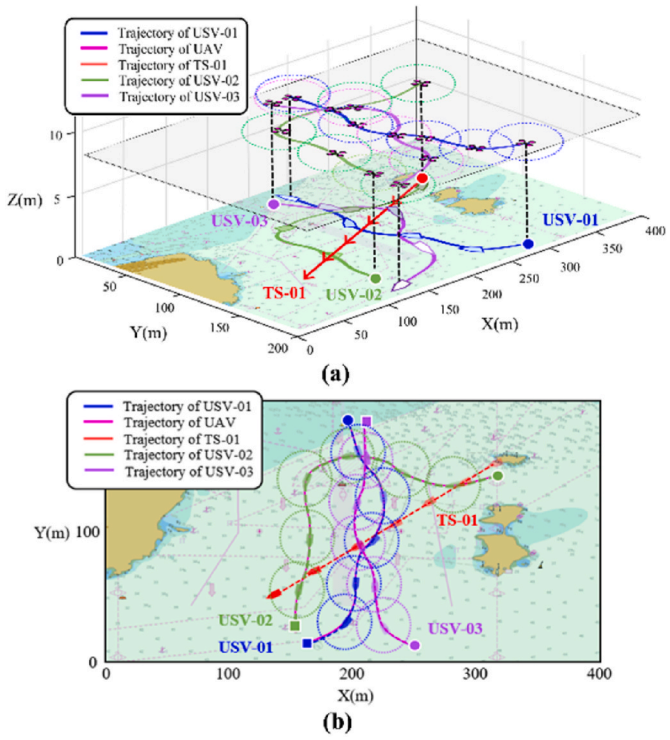


Fig. 18. The navigation collision avoidance trajectory of three UAV-USV formation and one TS in the situation of multiple ships encounter situation.

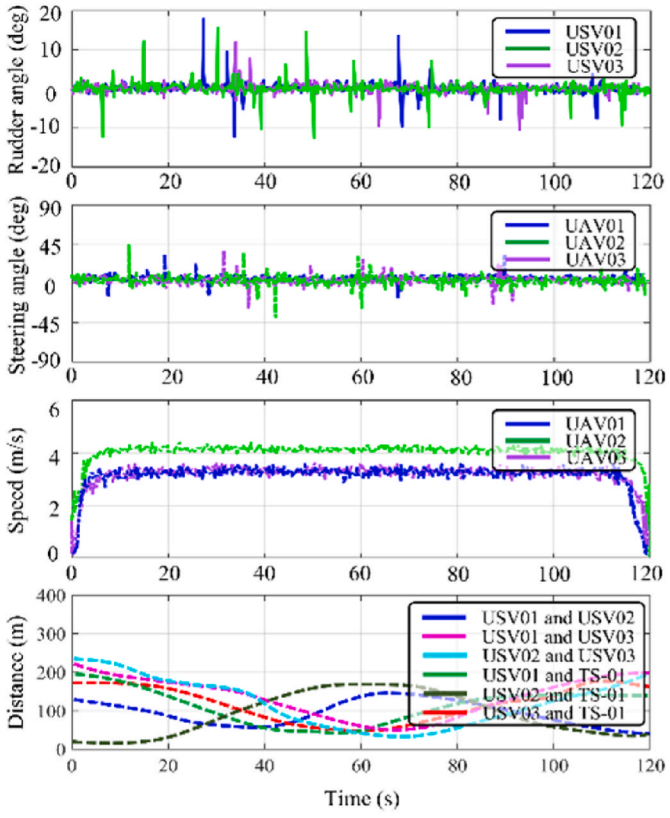


Fig. 19. The time-varying curves of the three USVs' rudder angle, three UAVs' steering angle and speed, and the distance between the two ships.

angle and speed, as well as the distance between the two ships during the encounter process, are presented in Fig. 17.

In addition, to increase the complexity of the experiment, a multi-formation collision avoidance test was also conducted involving three UAV-USV formations and one TS. The navigation process is illustrated in Fig. 18, and the detailed curve data is shown in Fig. 19. It is evident that even under highly complex scenarios, the proposed HCCAD strategy effectively ensures the navigational safety of the USV. Furthermore, across all encounter scenarios, the UAV's flight trajectory comprehensively covers the airspace both ahead of and behind the collision avoidance range, providing high-altitude visual perception for the USV, while maintaining a low collaboration error relative to the USV's path.

4.5. Comparation experiment

To evaluate the performance advantages of the UAV-USV collaboration collision avoidance method over traditional approaches, a comparative experimental analysis was conducted. The methods compared include the OB-MAPPO, OB-MAPPO-LiDAR relying solely on radar observations (without visual assistance from the UAV), MAPPO (Pu et al., 2025), MADDPG (Wang and Zhao, 2025), dynamic window approach-DQN (DWA-DQN) (Guan and Wang, 2023), and APF (Lyu and Yin, 2019). The adopted evaluation metrics include: generalization success rate (GSR) and average navigation time (ANT). The comparative results are shown in Table 3, Figs. 20 and 21. Based on the data, the following discussion can be developed.

Table 3
Comparision for GSR and ANT with different collision avoidance methods.

TS numbers	Generalization success rate (GSR)/Average navigation time (ANT)					
	OB-MAPPO	OB-MAPPO-LiDAR	MAPPO (Pu et al.)	MADDPG (Wang et al.)	DWA-DQN (Guan et al.)	APF (Lyu et al.)
1	**1.00 ± 0.0/ 200.6 ± 1.6**	1.00 ± 0.0/ 213.4 ± 1.7	1.00 ± 0.0/ 213.9 ± 2.1	1.00 ± 0.0/214.8 ± 1.9	1.00 ± 0.0/ 214.8 ± 2.3	1.00 ± 0.0/ 227.4 ± 3.1
2	**1.00 ± 0.0/ 212.3 ± 1.4**	1.00 ± 0.0/ 218.9 ± 1.9	1.00 ± 0.0/ 220.6 ± 3.7	1.00 ± 0.0/223.2 ± 3.6	1.00 ± 0.0/ 223.2 ± 3.6	1.00 ± 0.0/ 235.3 ± 3.5
3	**1.00 ± 0.0/ 218.5 ± 2.3**	1.00 ± 0.0/ 225.8 ± 2.4	0.99 ± 0.1/ 228.6 ± 2.9	1.00 ± 0.0/230.4 ± 2.8	0.95 ± 0.1/ 230.4 ± 3.7	0.91 ± 0.1/ 248.9 ± 4.2
4	**1.00 ± 0.0/ 224.8 ± 4.8**	0.96 ± 0.3/ 231.6 ± 4.8	0.97 ± 0.2/ 237.3 ± 5.0	0.94 ± 0.1/239.6 ± 5.4	0.91 ± 0.2/ 239.6 ± 4.9	0.87 ± 0.2/ 262.8 ± 5.3
5	**1.00 ± 0.0/ 228.9 ± 4.9**	0.95 ± 0.2/ 247.9 ± 4.7	0.96 ± 0.1/ 246.5 ± 5.6	0.92 ± 0.1/256.6 ± 5.9	0.90 ± 0.1/ 256.6 ± 5.8	0.86 ± 0.2/ 271.1 ± 5.9
6	**1.00 ± 0.0/ 237.5 ± 5.1**	0.92 ± 0.2/ 253.7 ± 5.9	0.94 ± 0.2/ 256.8 ± 5.9	0.89 ± 0.2/262.1 ± 6.1	0.86 ± 0.2/ 262.1 ± 6.2	0.84 ± 0.2/ 279.3 ± 6.7
7	**1.00 ± 0.0/ 237.9 ± 6.4**	0.90 ± 0.3/ 264.8 ± 6.6	0.89 ± 0.1/ 272.1 ± 6.2	0.89 ± 0.2/278.8 ± 6.8	0.81 ± 0.3/ 278.8 ± 7.3	0.77 ± 0.3/ 288.6 ± 7.1
8	**0.99 ± 0.1/ 244.1 ± 6.7**	0.88 ± 0.2/ 279.2 ± 8.8	0.85 ± 0.2/ 283.2 ± 7.7	0.84 ± 0.1/284.5 ± 7.4	0.80 ± 0.1/ 284.5 ± 7.6	0.71 ± 0.2/ 296.5 ± 7.8
9	**0.98 ± 0.2/ 248.6 ± 7.2**	0.88 ± 0.3/ 280.3 ± 8.8	0.83 ± 0.2/ 291.4 ± 8.0	0.82 ± 0.2/298.3 ± 8.1	0.75 ± 0.2/ 298.3 ± 8.9	0.65 ± 0.4/ 318.8 ± 8.9
10	**0.98 ± 0.2/ 251.3 ± 7.7**	0.85 ± 0.2/ 289.7 ± 9.0	0.83 ± 0.1/ 291.9 ± 8.5	0.79 ± 0.3/308.5 ± 8.7	0.71 ± 0.3/ 308.5 ± 9.4	0.59 ± 0.5/ 325.4 ± 9.8

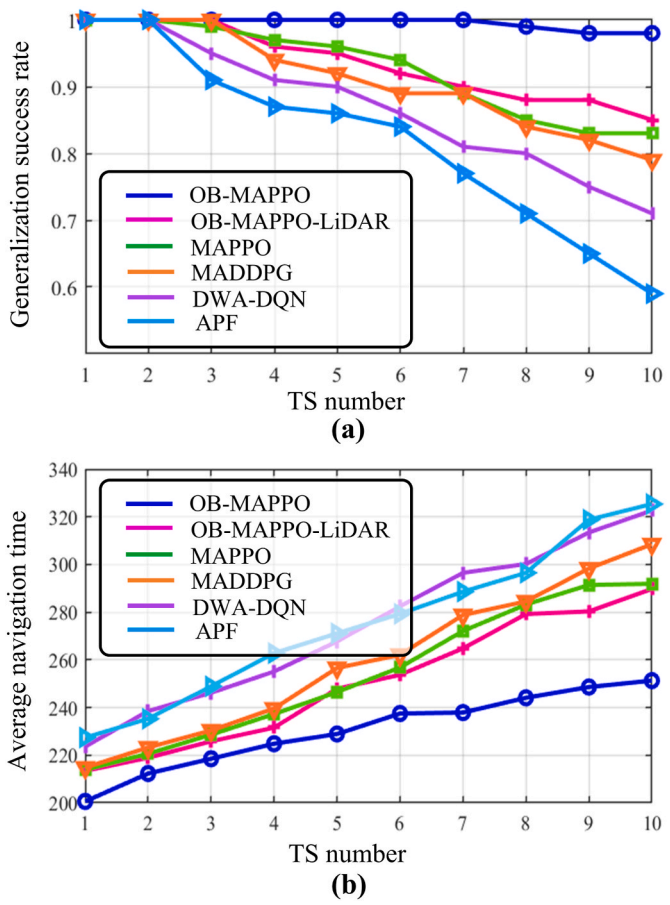


Fig. 20. The comparative results for GSR and ANT with different collision avoidance methods.

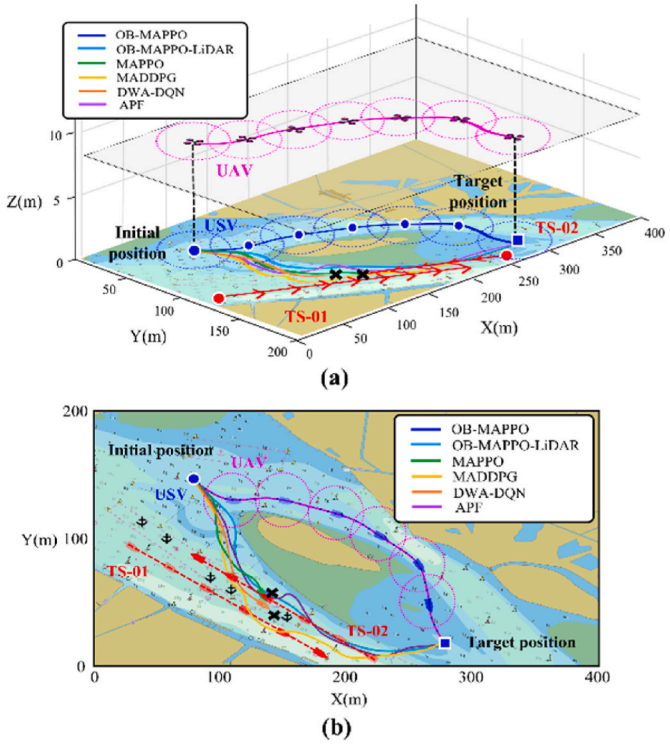


Fig. 21. Comparison of navigation trajectories of different collision avoidance methods in narrow waters.

Remark 3. It should be noted that, due to the inherent differences in model structures and optimization mechanisms across the compared algorithms, there is no unified parameter configuration that can guarantee optimal performance for all methods simultaneously. Therefore, in our experiments, we adopt algorithm-specific parameter settings and conduct reasonable tuning based on prior literature and publicly available implementations, ensuring that each method can achieve its representative performance. This design aims to provide a fair and objective comparison while reflecting the intrinsic performance differences among the evaluated approaches.

(1) As the number of dynamic TSs in the environment increases, the GSR of all collision avoidance strategies exhibits a corresponding decline, while the ANT shows a progressive rise. This is a commonly observed phenomenon (Table 3 and Fig. 20a). The OB-MAPPO algorithm leverages the dual sensing advantages of "LiDAR and vision" from the UAV-USV, enabling it to maintain the highest GSR across the entire scenario. Even as the scene complexity continuously increases (with the number of TSs rising from 1 to 10), no significant degradation in GSR is observed.

(2) Among the compared algorithms, the growth trend of the ANT for OB-MAPPO remains the most gradual as scenario complexity increases (Fig. 20b). This robustness can be attributed to the global visual information provided by UAV, which supports more efficient decision-making by USV, mitigates the risk of local optima convergence, and consequently improves overall navigation efficiency.

(3) Among all methods, DWA-DQN and APF exhibit relatively weaker performance. The APF guides the motion of the USV through the combined effect of attractive and repulsive forces. However, in narrow channel, positions with zero potential energy may arise, causing the USV to become trapped in a local optimum and unable to escape. Moreover, the approach is highly sensitive to obstacle geometry and layout, exacerbating the local minima issue in complex environments. Moreover, DWA-DQN inherits the inherent limitations of DQN, namely its inability to handle continuous action spaces and its tendency toward overestimation, which results in a rapid decline in its generalization ability as the number of TS increases.

(4) Fig. 21 provides a more intuitive illustration of the advantages of UAV-USV collaboration in addressing collision avoidance challenges within complex scenarios. During the initial navigation phase, all collision avoidance strategies except the present method chose to alter course to starboard to avoid the reef and proceed to the destination. This decision was based on the shorter distance and reduced travel time associated with a starboard alteration. However, this approach overlooks the presence of a greater number of anchored ships and passing ships on the starboard side. Equipped with UAV-based visual assistance providing a broader field of view, the OB-MAPPO algorithm opts instead for a port alteration, thereby avoiding dense obstacle areas and adopting a safer navigation strategy to reach the target position.

5. Conclusion

This study proposes an integrated HCCAD framework for UAV-USV collaboration based on the OB-MAPPO algorithm. In light of the heterogeneous nature of sea-air collaborative tasks, the proposed method incorporates the OB strategy and heterogeneous advantage decomposition mechanism at the algorithmic level. This approach effectively mitigates redundant information in parameter sharing arising from inconsistent action spaces among heterogeneous agents in conventional MAPPO, while also alleviating the problem of excessive variance in gradient estimation. In the network design, image and LiDAR observations are integrated to construct a joint state space, and a multimodal fusion network based on Swin Transformer is employed for feature

extraction. Furthermore, the reward function incorporates multiple critical factors, including ship domain, immediate danger situations, and collaborative errors. In the simulation, a series of comparative studies were conducted, including evaluations against MARL algorithms, classical encounter scenarios, complex multi-vessel situations, and alternative collision avoidance strategies. The results demonstrate the superior performance of our method in terms of convergence speed, CC, TT, TSR, GSR, and ANT. This study represents the first successful application of MARL to heterogeneous marine-aerial collaboration decision-making, breaking away from traditional distributed decision-making paradigms. It provides a novel theoretical foundation for collision-avoidance coordination among heterogeneous agents in complex, dynamic environments.

However, the proposed approach still has several limitations: (1) aerial obstacles are not considered; (2) the UAV is assumed to maintain a constant altitude during flight; (3) energy consumption issues between the UAV and USV are not addressed; and (4) The proposed framework partially relies on visual perception, which may be sensitive to real maritime conditions such as fog, sea spray, and lighting glare, potentially affecting multimodal fusion.

Future work will focus on enhancing robustness through uncertainty-aware multimodal fusion, improved visual perturbation modelling, and sim-to-real adaptation with real-world data. Further efforts will also explore more reliable cross-modal alignment strategies, with the aim of enabling stable and practical deployment of UAV-USV collaborative collision avoidance in real maritime environments.

CRediT authorship contribution statement

Zhewen Cui: Writing – original draft, Visualization, Methodology, Formal analysis. **Wei Guan:** Writing – review & editing, Methodology, Conceptualization. **Xianku Zhang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **C. Guedes Soares:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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